

$$\underline{F} = \gamma \left(\frac{\vec{F} \cdot \vec{v}}{c}, \vec{F} \right) = (F_0, F_x, F_y, F_z) = \gamma (F_0, F_x, F_y, F_z)$$

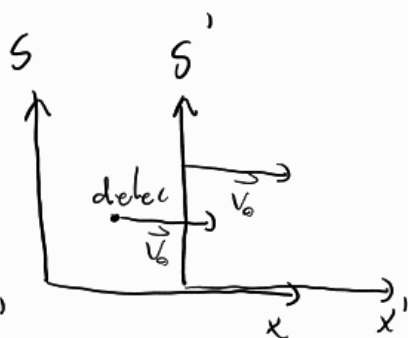
$$\vec{F} = e(\vec{E} + \vec{v} \times \vec{B})$$

$$F_x = \gamma F_x = \gamma_0 (F'_x + \beta_0 F'_0) = \gamma_0 \left(\gamma' F'_x + \beta_0 \gamma' \frac{\vec{F}' \cdot \vec{v}'}{c} \right)$$

$$(x = \gamma(x' + \beta t'))$$

$$\rightarrow F_x = \gamma_0 \gamma' \left(F'_x + \beta_0 \frac{\vec{F}' \cdot \vec{v}'}{c} \right)$$

↓
zadevatan
rel. gib. med S in S'



S' prijet na
delec
→ $\gamma' = 1$

$$F_y = \gamma F_y = \gamma' F'_y$$

$$F_z = \gamma F_z = \gamma' F'_z$$

$$\gamma_0 F_x = \gamma_0 \left(e E'_x + e (\vec{v} \times \vec{B})_x \right) = \gamma_0 \left(e E'_x + e (\vec{v} \times \vec{B})_x + \beta_0 \frac{\vec{F}' \cdot \vec{v}'}{c} \right)$$

ker delec prijet na S': $\vec{v}' = 0$

$$= v_y B_z - v_z B_y$$

ker se delec giblje vzdolž x osi

$$\Rightarrow \boxed{E_x = E'_x}$$

$$\gamma_0 F_y = \gamma_0 \left(e E'_y + e (\vec{v} \times \vec{B})_y \right) = \gamma' \left(e E'_y + e (\vec{v}' \times \vec{B}')_y \right)$$

$$= v_x B_z - v_z B_x$$

$$\boxed{E'_y = \gamma_0 (E_y - v_0 B_z), \quad E'_z = \gamma_0 (E_z + v_0 B_y)}$$

obratni transf.

$$\begin{cases} \Sigma_y = \gamma_0 (\Sigma'_y + v_0 B'_z) \\ \Sigma_z = \gamma_0 (\Sigma'_z - v_0 B'_y) \end{cases} \quad (2)$$

① vstavimo v ②:

$$\begin{aligned} \Sigma_y &= \gamma_0 (\gamma_0 (\Sigma_y - v_0 B_z) + v_0 B'_z) = \\ &= \gamma_0^2 \Sigma_y - v_0 \gamma_0^2 B_z + \gamma v_0 B'_z = \end{aligned}$$

$$\begin{cases} B'_z = \gamma_0 (B_z - \frac{v_0}{c^2} \Sigma_y) \\ B'_y = \gamma_0 (B_y - \frac{v_0}{c^2} \Sigma_z) \end{cases}$$

B_x :

- dajmo, da je v S mag. polje le vzdolž x

$\Rightarrow$ v S' je lahko nen ničelno le B_x

$$\vec{v} = (v, 0, 0) \Rightarrow \text{v S'}: \vec{v}' = (0, 0, v)$$

$$\Rightarrow \gamma F_y = \gamma (e(\vec{v} \times \vec{B})_y) = \gamma e v B_x = \gamma' F'_y = \gamma' e v' B'_x$$

$$\Rightarrow \gamma' v' B'_x = \gamma v B_x$$

$$\underline{v} = (\gamma c, \gamma \vec{v}) \Rightarrow \gamma' v' = \gamma v \Rightarrow \boxed{B'_x = B_x}$$

$$\underline{L} = \begin{pmatrix} \gamma - \beta \gamma & 0 \\ -\beta \gamma & \gamma \end{pmatrix} \quad \text{v } \mathcal{O}_1$$

$\rightarrow$ zdaj znamo transf. $\vec{\Sigma}, \vec{B} \leftrightarrow \vec{\Sigma}', \vec{B}'$

$\rightarrow$ zložimo jih v EM tenzor:

$$\underline{F} = \begin{bmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & B_z & -B_y \\ -E_y/c & -B_z & 0 & B_x \\ -E_z/c & B_y & -B_x & 0 \end{bmatrix}$$

Transformirana se z LT:

$$\boxed{F'_{\mu\nu} = L_{\mu\rho} L_{\nu\sigma} F_{\rho\sigma}}$$

$$\text{oz. } \underline{F}' = \underline{L} \underline{L} \underline{F}$$

